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Non-Invasive Hemoglobin prediction: A machine learning approach

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Abstract—The global health crisis, which is characterized by low hemoglobin levels, is known as anemia and affects more than three billion individuals worldwide and is a cause of significant morbidity and mortality. The diagnostic techniques most commonly used in classical diagnosis, which are typically invasive blood tests, are often inconvenient, painful, and resource-intensive, and large-scale screening and continued surveillance is virtually impossible, particularly in low-resource settings. The present review looks at the new direction in non-invasive hemoglobin and anemia forecasting based on the ground-breaking potential of machine learning (ML) algorithms. This paper knowledge regarding a variety of 25 academic researches, the types of non-invasive sensing modalities studied, the scope of the diverse ML algorithms that have been applied, and the performance metrics achieved with them. It highlights how these technologies can transform the face of anemia detection by offering real- time, painless, and low-cost modalities of diagnosing anemia. Moreover, the review identifies such significant issues as diversity of datasets, clinical validation, and algorithmic bias because it identifies potential future research opportunities such as federated learning and smartphone-based screening in order to facilitate the successful implementation and delivery of such innovative approaches.

Keywords: Hemoglobin, Anemia, Non- invasive Detection, Machine Learning, Deep Learning, Healthcare AI, Photoplethysmography.

I. Introduction

Anemia, characterized by a defect in the quantity of red blood cells or the level of hemoglobin (Hb) in them, compromises the blood's oxygen-carrying ability for conciseness, resulting in tiredness, weakness, and organ failure and even death in advanced stages. Anemia is an overarching public health concern worldwide, most affecting women and children in low and middle-income nations. Its prevalence highlights the pressing need for affordable, effective, and scalable diagnostic equipment [1], [2].

Traditionally, invasive blood sampling e.g. complete blood count (CBC) or spectrophotometry have been used to measure hemoglobin concentration. Even though extremely accurate, they are inconvenient to use regularly because of venipuncture, laboratory testing, skilled staff, and a central location and are, therefore, not applicable to convenient monitoring and replication in peripheral or resource-rich environments [1], [3]. Risk associated with invasive testing which is accompanied by pain and infection also discourage regular screening resulting in late diagnosis and treatment.

The emergence of machine learning (ML) and artificial intelligence (AI) has introduced new opportunities in healthcare with new prospects of machine-diagnosis and personalized medicine. ML algorithms are more robust in discovering weak signals in massive, noisy data and thus is most applicable to the situation of decoding physiological signals and predicting health values through non-invasive measurements [4], [5]. Such a non-invasive sensing technologies and state of the art computational model's interaction has enormous potential of transforming anemia detection into an active, patient-based process as opposed to being a reactive, clinic-based process.

This review article will focus on trying to provide the entire picture of the current state of research on the non-invasive non-invasive hemoglobin prediction and anemia prediction based on machine learning techniques. Specifically, it has combined findings of 25 studies, analyzing various non-invasive modalities, the machine learning algorithms employed, their performance as reported, and the challenges and limitations. By synthesizing the state of knowledge in the described fashion, the purpose of the review is to: provide guidelines on how to proceed with future work so as to accelerate the development and clinical practice of these new diagnostic tools.

II. Background And Related Work

Measurement of proper hemoglobin concentration is essential in measuring and therapy of anemia. The conventional methods, such as complete blood count (CBC) and spectrophotometry, involve the collection of blood samples, which are examined in the laboratory. Spectrophotometry, such as that, computes the concentration using light that is absorbed by hemoglobin derivatives, which is quite accurate and reliable in terms of accuracy and reliability [3]. The invasive nature of these methods, time requirements and complexity of their logistical aspects, despite their diagnostic gold standard and the need to develop alternatives that are non-invasive in nature [1].

The goal of non-invasive hemoglobin estimation methods is to avoid the shortcomings of conventional methods by examining physiological signals acquired without penetration of the skin surface. Photoplethysmography (PPG) is a highly

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visible non- invasive imaging modality that is used to measure volumetric blood circulation changes by monitoring fluctuations in light absorption or reflection from tissue [3], [6], [7]. The other methods involve near-infrared (NIR) spectroscopy and camera imaging of body regions such as the conjunctiva or fingertip [8], [9]. Some of the initial research into anemia detection non-invasively used simpler mathematical models or statistical relationships between optical signals and hemoglobin concentrations. For example, some techniques separated the RGB components of the input photos and used an algorithm to determine and estimate the hemoglobin content. [9]. Tamir et al. used thresholding on the red and green picture components to determine whether a user had anemia, however this approach was unsophisticated and might not have detected the complexity of image input changes. [9]. Khan et al. developed a hardware-based prediction technique that combines a linear regression model specifically designed for blood flow-based prediction with an FPGA-based architecture. Despite being inventive, its scalability and availability are limited by its dependency on hardware [9]. Using conjunctival pictures, Collings et al. calculated the erythema index (EI) and investigated its correlation with hemoglobin content. The prediction performance may be limited if only one feature is used. [9]. Chen et al. subsequently used modified Kalman Filtering (KF) to further apply EI, and through penalty regression, they were able to achieve good predictive accuracy on both low-risk and high-risk data. The feature set that was chosen still placed limitations on this approach, although [9]. After preprocessing for brightness leveling, Anggraeni et al. created a linear regression model using self-collected conjunctival images of expectant mothers. The model's performance was somewhat constrained by the preprocessing stages' standards. [9]. These studies used basic mathematical models to diagnose anemia while integrating several computer vision techniques. The majority of these methods relied on specific preprocessing procedures and limited feature sets, which may not work well with photos of different quality levels. [9].

One significant advancement has been the application of machine learning to non-invasive hemoglobin estimation. Early uses of ML for Hb estimation included several regression methods to estimate hemoglobin levels from PPG signals and other physiological characteristics .In order to demonstrate the ability of these models to estimate hemoglobin non-invasively, Kavsaog lu et al. (2015) examined a number of machine learning regression techniques, including Classification and Regression Trees (CART), Least Squares Regression (LSR), Generalized Linear Regression (GLR), Multivariate Linear Regression (MVLR), Partial Least Squares Regression (PLSR),

Generalized Regression Neural Network (GRNN), Multilayer Perceptron (MLP), and Support Vector Regression (SVR) [5]. Previous reviews have tended to concentrate on a particular element of non-invasive diagnostics or the application of machine learning to healthcare. Yet, there is a need for an overall synthesis that addresses, in particular, the confluence of non-invasive hemoglobin and prediction of anemia with varied machine learning approaches as well as outlining sensors, data acquisition, and a general array of challenges and future avenues. This review aims to fill this gap by providing a structured analysis of the current research landscape, drawing from a curated set of 25 relevant academic papers.

TABLE I. COMPARISON OF INVASIVE VS. NON-INVASIVE HEMOGLOBIN ESTIMATION TECHNIQUES

Feature	Invasive Methods (e.g., CBC, Spectrophotometry)	Non-Invasive Methods (e.g., PPG, Imaging)
Procedure	Requires blood sample (venipuncture)	No blood sample required (e.g., finger probe, camera)
Pain/Discomfort	Painful, potential for discomfort	Painless, comfortable
Infection Risk	Present (due to skin puncture)	Minimal to none
Real-time Monitoring	Limited, typically intermittent	Potential for continuous or real-time monitoring
Equipment	Specialized lab equipment, trained personnel	Portable devices (smartphones, embedded systems), less specialized training
Cost per Test	Generally higher (lab fees, consumables)	Potentially lower (device cost, no consumables)
Accessibility	Limited by lab infrastructure	High, suitable for remote and home use
Accuracy	High (gold standard)	Variable, improving with ML/DL advancements
Data Acquisition	Manual blood collection	Automated sensor data capture

III. MACHINE LEARNING TECHNIQUES USED

The utilization of machine learning (ML) and deep learning (DL) methods is at the heart of developing non-invasive hemoglobin and anemia forecasting. Researchers have experimented with a wide range of models, from classical statistical regression to highly involved neural networks, to derive useful information from physiological signals.

Typically used models are Support Vector Machines (SVM) and its variations such as Support Vector Regression (SVR), which can be used for both classification and regression problems by determining best hyperplanes in high dimensions [5], [14]–[22]. Random Forest (RF) and K-Nearest Neighbors (KNN) are also used for their stability and capacity to address intricate, non-linear relationships within data. Artificial Neural Networks (ANNs), such as Multilayer Perceptron (MLP) and Generalized Regression Neural Networks (GRNN), have been used to capture the complex mappings between non-invasive signals and hemoglobin concentrations [5].

Recent deep learning architecture, including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), has come to the fore because they can automatically learn hierarchical features from raw image or signal data, eliminating the necessity for feature engineering. For example, Chen et al. (2024) and Lin et al. (2024) constructed deep learning models for real-time non-invasive prediction of hemoglobin from smartphone images and body part im- ages, respectively, with high accuracy and F1-scores [9], [10]. C¸ uvadar and Yılmaz (2023) also used deep learning to estimate hemoglobin from conjunctival images [8]. Yılmaz et al. (2022) proposed a novel hybrid deep learning approach for highly accurate non-invasive hemoglobin estimation [11].

Training and test procedures generally mean dividing datasets into training, validation, and test sets for the purpose

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of model generalization. Performance is measured by indicators such as Mean Absolute Error (MAE), Mean Square Error (MSE), Root Mean Square Error (RMSE), R-squared (R2) coefficient of determination, Mean Absolute Percentage Error (MAPE), and Index of Agreement (IA) [5]. For instance, Pinto et al. (2020) have reported R2 = 0.964 for their five-wavelength PPG system [3]. Golap et al. (2021) reported MAE of ±0.304 and ±0.324 for hemoglobin and glucose, respectively, with a multigene genetic programming (MGGP)-based model [7]. Lin et al. (2024) reported as high as 0.716 for prediction accuracy and 0.788 for F1-score for their deep learning model [9].

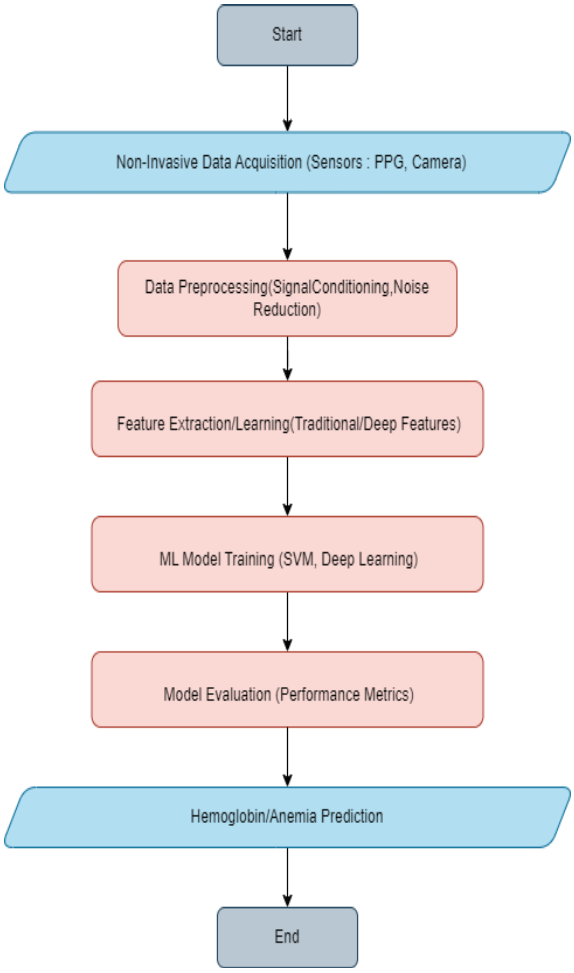


Fig. 1. Flowchart showing Data Collection and Prediction in ML Pipeline for Anemia Prediction

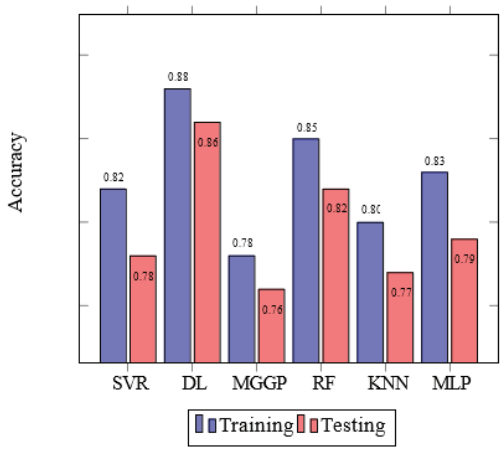


Fig. 2. Bar Graph of Accuracy vs. Algorithm from Reviewed Studies

Ensemble models that use several individual models working together to enhance predictive performance overall are also being investigated. These can help reduce the disadvantage of overfitting and generalize better across various datasets.

Yet, difficulties remain, such as the potential overfitting to individual datasets, intrinsic biases (e.g., with respect to skin color, age, or gender) if not properly accounted for in data acquisition, and making models generalizable across various populations and device types [5]. The diversity and quality of training data are most important for creating robust and clinically feasible non-invasive prediction models.

IV. Sensors And Data Acquisition Systems

The effectiveness of non-invasive hemoglobin and anemia predictive models relies necessarily on the quality and nature of data collected using different sensors. The overall objective is to measure physiological signals related to hemoglobin levels without actual blood sampling.

Pulse oximeters are traditional devices used to measure oxygen saturation and are based on the principle of photoplethysmography and are a fundamental technology of non-invasive hemoglobin. PPG signals, which are obtained by illuminating the tissue (usually the tip of the fingertip or ear lobe) with light and measuring the change in light absorption caused by pulsatile blood flow, provide a detailed signal with regards to changes in blood volume [3], [6], [7]. To detect these signals investigators have developed special finger probes with multiple LED light sources of different wavelengths (e.g. 670 nm, 770 nm, 810 nm, 850 nm, 950 nm) and photodetectors, which they find more accurate with multiple wavelengths [3].

The pervasive use of smartphones has made them a forceful platform for non-invasive diagnosis. Smartphone cameras are capable of recording RGB images or videos of body tissues such as the conjunctiva or fingertip, which, with the help of computer vision and deep learning algorithms, are processed to estimate hemoglobin concentration [8]–[10]. The camera- based method has an enormous potential in accessibility and the likelihood of mass screening in rural communities [1].

Data modalities are mainly PPG waveforms, spectroscopy data, and RGB imaging. The PPG signals tend to be analyzed for characteristic features, such as derivatives and Fourier analysis, to obtain applicable parameters for machine learning models [5], [7]. Spectroscopy, which involves the study of how light interacts with biological tissue of a varied wavelength range of wave- lengths, provides a high level of accuracy of information about chromophore concentrations, including hemoglobin.

The potential of real-time data acquisition is a challenge in a number of ways. Due to the motion of the patient during data collection, motion artifacts can be experienced, which entails noise and severely affects signal quality affecting prediction accuracy [13]. The reliability and consistency of optical measurements are also affected by inconsistencies in ambient light intensity and skin color and, hence, vigorous preprocessing techniques and divergent data sets are necessary to affect these issues [9]. Dataset generating equipment is more likely to have capability to reduce such artifacts, such as removing start and endpoints of recordings where movement is prevalent [13].

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TABLE II. SUMMARY OF REVIEWED STUDIES: AUTHORS, ML MODELS, ACCURACY, DATASET, MODALITY USED

SN	Author(s)	ML Model(s)	Performance Metric (Value)	Dataset Characteristics	Modality Used
1	Chen et al. [10]	Deep Learning	Accuracy: 0.716, F1-score: 0.788	101 patients (prospective)	Smartphone Imaging
2	C¸uvadar & Yılmaz [8]	Deep Learning	Not specified	Conjunctival images	Conjunctival Imaging
3	Yılmaz et al. [11]	Deep Learning	High accuracy	Not specified	Not specified
4	Hasan et al. [1]	Not specified (Review)	N/A	N/A	Mobile Phone Environment
5	Chen et al. [6]	Not specified (System)	Not specified	Not specified	Four-Wavelength PPG
6	Vijay & Mehfooza [2]	Data-Driven Approach	Not specified	Not specified	Not specified
7	Abuzairi et al. [12]	Machine Learning	Not specified	Hb blood value + PPG signal	PPG
8	Dauvin et al. [4]	Machine Learning	Not specified	ICU patients	Not specified
9	Berry et al. [23]	N/A (Statistical)	N/A	N/A	N/A
10	Lin et al. [9]	Deep Learning	Accuracy: 0.716, F1-score: 0.788	101 patients (retro & prospective)	Body Parts Images
11	Burrell [24]	N/A (Quasi-experimental)	N/A	Rural population	N/A
12	Mosaddequr & Rahman [13]	Regression Model	Accuracy: 0.7850 mmol/L	50 people (age-diverse)	Reflection PPG
13	Golap et al. [7]	MGGP	MAE: ±0.304 (Hb), ±0.324 (Glucose)	Fingertip video (PPG)	Fingertip Video
14	Pinto et al. [3]	Algorithm (Regression)	R2: 0.964, MAE: 0.3499 g/dL	8 subjects	PPG with embedded system
15	Berry et al. [25]	N/A (Statistical)	N/A	N/A	N/A
16	Li et al. [14]	Proximal SVM	Not specified	Class/Reg datasets	N/A
17	Zhao et al. [15]	Multi-output SVR	Not specified	Concrete data	N/A
18	Kong et al. [16]	RSVM	Not specified	Wind speed data	N/A
19	Xu et al. [17]	Weighted Quantile Reg	Not specified	Benchmark datasets	N/A
20	Maalouf & Homouz [18]	KRR	Not specified	Six datasets	N/A
21	Dai et al. [19]	Multiwavelet SVR	Not specified	Structural limit states	N/A
22	Hu & Zheng [20]	SVR	Not specified	Dataset with outliers	N/A
23	Wang et al. [21]	Multiple-SVR	Not specified	Not specified	N/A
24	Kavsaog˘lu et al. [5]	SVR, CART, LSR, GLR, MVLR, PLSR, GRNN, MLP	Best with SVR (RELIEFF)	33 people, 40 PPG features + demographics	PPG signals
25	Ortun˜o et al. [22]	ML & Math Regression	Not specified	Multiple sequence alignments	N/A

V. CHALLENGES AND LIMITATIONS

In spite of impressive advances in non-invasive hemoglobin prediction and anemia prediction with machine learning, there are a number of thorny issues and limitations related to the application of this method to a broad range of patients and clinical settings.

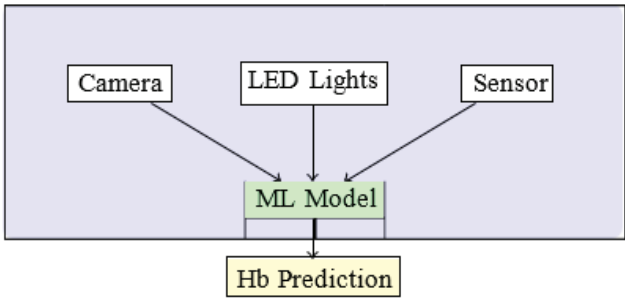


Fig. 3. Diagram showing Non-Invasive Sensor Setup (Camera + ML Model)

Dataset Diversity and Volume is one of the biggest hindrances to clinical translation. A large number of studies use relatively small cohorts, usually based on homogenous populations and that seriously reduces the extrapolation of trained models to larger groups of the population [3], [13]. Systematic algorithmic bias can be caused by the absence of diversity in datasets, especially in terms of differences in skin pigmentation, age categories, gender balance, and comorbidities underlying an individual. The proposed models that are primarily trained on data of a certain group of demographics can have poor performance or less accuracy in other underrepresented populations, thus worsening current health inequities [13]. Although age diversity has been tried to be integrated in some studies [13], the attainment of representation of all the significant demographic variables is a big challenge.

Another significant challenge is Clinical Validation and Regulatory Approval. Despite the laboratory-based investigations showing promising accuracy rates, clinical trials that are rigorous, multi-centric and are conducted in real-clinical settings, in comparison to gold-standard invasive capabilities and in different patient populations are necessary to confirm the effectiveness of such non-invasive devices [9]. The changes between research prototypes and clinically approved devices are associated with a lengthy period of validation studies, which will be needed to prove safety, efficacy, and reliability when exposed to different clinical conditions.

This has major challenges to comparative analysis and clinical implementation of different devices and methodologies because they are not standardized. Various research teams use different types of sensors, light

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wavelengths, data capture schemes and preprocess, which makes direct comparison between studies complicated and prevents creating universal models and effective benchmarks. As an example, some researchers make use of multi-wavelength PPG system [3], [6], whereas others use smartphone cameras [10], which brings confusion to the discipline.

Environmental and Technical Limits are also very problematic. The movement of the patient during the recording procedure can cause motion artifacts that bring significant noise and deteriorate the quality of signals, which directly affects the quality of prediction [13]. Ambient variations, differences in skin pigmentation, and personal physiological changes can also contribute to the variability and lack of consistency of optical data, which is why good preprocessors are required and many different training sets will have to be used to take these factors into account [9].

There are ethical and Safety Concerns of privacy of the data and accuracy of prediction. Sensitive physiological data are gathered by non-invasive devices and this calls on strict data protection and privacy measures. Misdiagnosis of anemia (false positive or false negative) can cause severe clinical outcomes, for example, unnecessary medical treatment, or delay in essential treatment. These devices should also be able to achieve exceptionally high levels of accuracy and reliability to guarantee the safety of patients and also to guarantee clinical confidence.

Moreover, the use of particular preprocessing stages and small sets of features in certain previous mathematical models have also indicated the serious drawbacks of treating different image qualities and different clinical conditions [9]. The effects of these limitations highlight the necessity of more robust and generalizable methods that could be effectively used in various imaging conditions and populations of patients.

TABLE III. DISTRIBUTION OF MOST FREQUENTLY MENTIONED CHALLENGES

Challenge Category	Frequency (%)
Dataset Diversity & Volume	40
Clinical Validation	25
Algorithmic Bias	20
Device Standardization	15

VI. FUTURE RESEARCH DIRECTIONS

The current area of predicting non-invasive hemoglobin and anemia with the help of machine learning is actively developing, and there are a range of potential future directions of this research that may help overcome the existing drawbacks and speed up the process of clinical implementation.

The creation of larger, more diverse, and clinically validated datasets is one of such important directions. The process of the collection of the data on the multi-centers in the future must consider all the features of the demographic variables including different skin colors, ages, genders, and diseases to give a model its strength and reduce bias [12]. The building and trading of high-quality specialist datasets become more and more regarded as a prestigious scholarly input that enhances reproducibility and accelerates the pace of innovation [12], [13].

The incorporation of federated learning may provide a solution to data privacy while making it possible to train stronger models. Federated learning enables models to be trained on decentralized data at local institutions without their raw patient data being shared, thus maintaining privacy and complying with regulations [1].

Increased smartphones-based screening developments are essential, especially for distant and underserved populations. Expanding the precision and reliability of smartphone camera-based systems, perhaps by using sophisticated image processing methods and modeling powerful deep learning models, could allow for generalized low-cost anemia screening programs [1], [10].

Interoperability with Electronic Health Record (EHR) systems is a major advance towards clinical usefulness. Smooth integration would enable non-invasive hemoglobin information to be automatically captured, monitored, and correlated with other patient health data, making clinical decision making and ongoing patient monitoring possible in a more effective manner.

There can be no overestimation of the requirement for multicenter clinical trials. Future studies need to shift their focus from laboratory environments to robustly test non-invasive devices against gold-standard measures within varied clinical settings, evaluating their performance, usability, and effect on patient outcomes.

Lastly, the use of explainable AI (XAI) methods will be critical in building trust and transparency in ML-based diagnostic applications. Clinicians can make sense of outcomes, detect possible biases, and maintain patient safety, particularly in high-stakes diagnostic situations, if they know how these sophisticated models reach their conclusions.

VII. CONCLUSION

The global anemia clinical challenge needs non-invasive and less expensive methods of diagnosis by the traditional methods. This literature review has revealed the enormous potential of machine learning-based prediction of non-invasive hemoglobin and anemia, which is based on 25 unavoidable scientific papers. The present research landscape is characterized by a strong focus on the non-invasive technologies like photoplethysmography and smartphone imaging, paired with the latest machine learning and deep learning algorithms.

Enormous steps have been made in developing models which can estimate hemoglobin density with more accuracy, making it pain free, real time and possibly continuous monitoring possible. The data-driven publications boom can be interpreted as a growing interest in high quality datasets as the key to driving this technology to the next level. However, major issues remain, including the need to have more heterogenous and larger data sets, appropriate clinical validation in a wide variety of populations, standardisation of devices, and the problem of data privacy and algorithm bias.

Future research directions include the collective efforts in data collection, the inclusion of privacy-preserving solutions like federated learning, and the integration of such technologies in existing healthcare institutions. The continued investment into multi-center clinical trials and development of explainable AI models will be instrumental in achieving trust and successful and safe applications of non-invasive anemia prediction machines. The current developments in this field have the capability of transforming the management of anemia

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by enhancing early diagnosis and prevention of the condition in most parts of the world.

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